

FinRelief AI: AI-Powered Debt Relief & Financial Recovery Platform

Milestone 5: Deployment

Milestone 5 focuses on deploying the FinRelief AI application locally to verify its functionality and then deploying it publicly using Render (Backend) and Vercel (Frontend). This ensures that users can securely access the application through a web browser while interacting with AI-powered financial analysis and debt settlement recommendations.

Activity 5.1: Preparing the Application for Local Deployment

Set Up a Virtual Environment

- Open Command Prompt or VS Code Terminal.
- Navigate to the backend folder.

```
cd backend
```

- Create and activate a virtual environment.

```
python -m venv venv
```

```
Windows  
venv\Scripts\activate
```

- Install all required dependencies.

```
pip install -r requirements.txt
```

Configure Environment Variables

- Create a .env file inside the backend folder.
- Add the following:

```
GEMINI_API_KEY=your_google_gemini_api_key  
SECRET_KEY=your_secret_key  
DATABASE_URL=sqlite:///finance.db
```

The application automatically loads these variables using python-dotenv.

Activity 5.2: Local Testing and Verification

Start the FastAPI Backend

```
cd backend  
uvicorn main:app --reload
```

The backend runs at

```
http://127.0.0.1:8000
```

Start the React Frontend

- Open another terminal.

```
cd frontend  
npm install  
npm run dev
```

The frontend runs at

```
http://localhost:5173
```

Verify the Application

Test the following features:

- User Registration
- User Login
- Loan Management
- Financial Health Analysis
- AI Debt Settlement Recommendation
- Dashboard
- User Profile

Verify that all API requests are processed successfully and data is stored in the SQLite database.

Activity 5.3: Public Deployment

Backend Deployment using Render

Deploy the FastAPI backend on Render.

- Build Command

```
pip install -r requirements.txt
```

- Start Command

```
uvicorn main:app --host 0.0.0.0 --port $PORT
```

After deployment, Render provides a public backend URL such as:

```
https://finrelief-ai-backend.onrender.com
```

Frontend Deployment using Vercel

Deploy the React frontend on Vercel.

- Framework: Vite
- Build Command: npm run build
- Output Directory: dist

After deployment, Vercel provides a public frontend URL such as:

```
https://finrelief-ai.vercel.app
```

Connect Frontend with Backend

- Create a .env file in the frontend directory.

```
VITE_API_URL=https://finrelief-ai-backend.onrender.com
```

Use this variable in Axios or Fetch requests to communicate with the backend.

Final Verification

Verify the deployed application by testing:

- User Registration
- Login Authentication
- Loan Management
- Financial Analysis
- AI Recommendations
- Dashboard
- Profile Management

Ensure all frontend requests successfully communicate with the backend APIs.

Exploring the Website Pages

Home Page

Description: The home page serves as the main landing page of FinRelief AI. It provides an overview of the platform, navigation options, and allows users to access financial management services.

 **FinRelief AI**

LoginRegister

AI-Powered Debt Management

Manage your loans, analyze your financial health, and receive intelligent AI-powered settlement recommendations—all in one place.

Get Started

 **Financial Analysis**
Calculate EMI ratio, debt ratio, health score, monthly surplus, and stress level.

 **AI Recommendation**
Receive AI-generated debt settlement plans tailored to your financial situation.

 **Loan Management**
Add, update, and monitor all your loans in one secure dashboard.

Login & Registration

Description: Users can securely register and log in using JWT-based authentication. Passwords are encrypted to ensure account security.

Register

Full Name

Email

Password

Register

Already have an account? [Login](#)

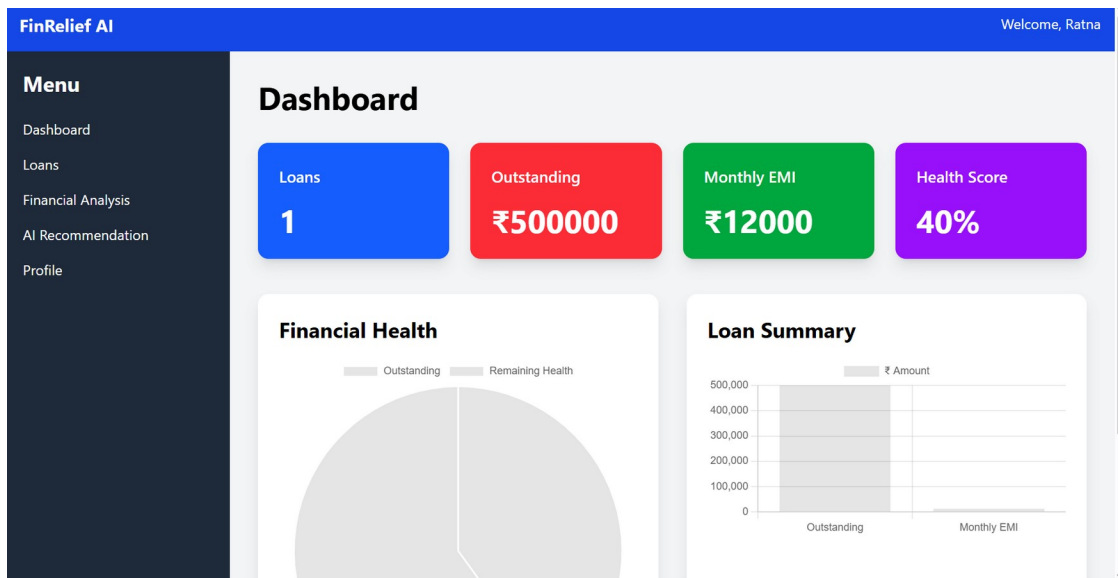
Login

Login

Don't have an account? [Register](#)

Dashboard

Description: Displays a summary of active loans, repayment status, financial health score, and AI-generated financial insights through charts and reports.



Loan Management

Description: Users can add, edit, delete, and view loan details, including lender information, interest rate, EMI amount, and repayment schedule.

FinRelief AI

Welcome, Ratna

Menu
Dashboard
Loans
Financial Analysis
AI Recommendation
Profile

Loan Management

+ Add Loan

Search Loan...

Loan Name	Lender	Outstanding	EMI	Interest	Overdue	Status	Actions
Home Loan	SBI	₹500000	₹12000	8%	6 Months	Active	EditDelete

Menu
Dashboard
Loans
Financial Analysis
AI Recommendation
Profile

Add Loan

Loan Name

Lender

Outstanding Amount

Monthly EMI

Interest Rate (%)

Overdue Months

Save Loan

Financial Analysis

Description: Analyzes income, expenses, and liabilities to calculate the user's financial health score and identify areas for improvement.

Financial Analysis

AI Debt Settlement Recommendation

Description: Uses Google Gemini AI to generate personalized debt repayment strategies, settlement suggestions, and financial recovery recommendations.

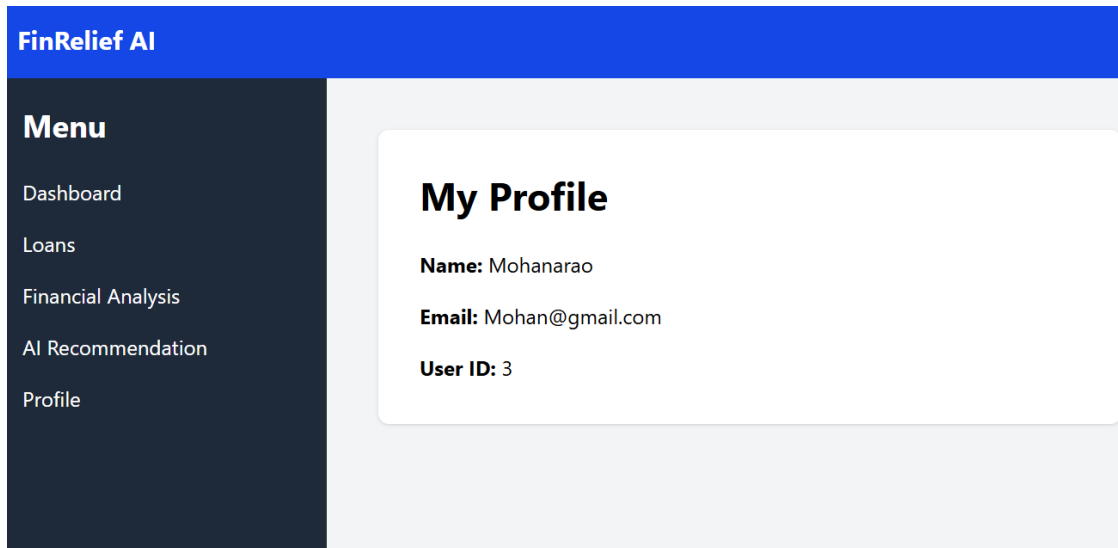
FinRelief AI

Menu
Dashboard
Loans
Financial Analysis
AI Recommendation
Profile

 **AI Debt Recovery Assistant**

Profile Page

Description: Allows users to update personal information, change passwords, and manage account settings securely.



Conclusion

FinRelief AI is an AI-powered debt management and financial recovery platform developed using React.js, FastAPI, SQLite, SQLAlchemy, and Google Gemini AI. The application enables users to manage loans, monitor financial health, and receive intelligent debt settlement recommendations. The project demonstrates the practical application of Generative AI in personal finance while providing a scalable, secure, and user-friendly solution. Future enhancements include cloud database integration, email notifications, multilingual support, credit score integration, and mobile application development.